

Customer Churn Prediction System Documentation

Project Title

Customer Churn Prediction System using Machine Learning

Project Objective

The objective of this project is to develop a Machine Learning model that predicts whether a telecom customer is likely to churn (leave the company) or not. This helps telecom companies identify high-risk customers and take preventive actions to improve customer retention.

Phase 1: Exploratory Data Analysis (EDA)

Objective

Understand the dataset and identify important patterns before building the machine learning model.

Steps Performed

- Loaded the Telco Customer Churn dataset using Pandas.
- Displayed the first five rows of the dataset.
- Checked dataset shape and data types.
- Identified missing values.
- Analyzed numerical and categorical features.
- Created visualizations to understand customer behavior.

Visualizations

- Churn Distribution
- Contract Type Distribution
- Monthly Charges Distribution
- Tenure Distribution
- Correlation Heatmap

Key Observations

- Customers with **Month-to-Month** contracts are more likely to churn.
- Customers with **higher Monthly Charges** have a higher churn rate.
- Customers with **longer tenure** are less likely to leave the company.
- **Fiber Optic** users show a higher churn rate than DSL users.
- Two-year contract customers have the lowest churn rate.

Phase 2: Data Preprocessing

Objective

Prepare the dataset for machine learning by cleaning and transforming the data.

Steps Performed

- Removed the unnecessary customerID column.
- Converted TotalCharges to numeric format.
- Filled missing values using the median.
- Encoded categorical variables using Label Encoding.
- Scaled numerical features using StandardScaler.
- Split the dataset into Training (80%) and Testing (20%).

Output

The dataset was transformed into a machine-learning-ready format.

Phase 3: Model Development

Objective

Train multiple machine learning models and compare their performance.

Models Used

- Logistic Regression
- Random Forest Classifier
- Gradient Boosting Classifier

Process

- Trained all three models using the training dataset.
- Generated predictions on the testing dataset.
- Compared their performance using evaluation metrics.

Phase 4: Model Evaluation

Objective

Evaluate the trained models and select the best-performing model.

Evaluation Metrics

- Accuracy
- Precision
- Recall
- F1-Score

- ROC-AUC Score
- Confusion Matrix

Model Comparison

Model	Accuracy	Precision	Recall	F1 Score
Logistic Regression	80%	69%	56%	62%
Random Forest	79%	67%	53%	59%
Gradient Boosting	81%	71%	58%	64%

Conclusion

Gradient Boosting achieved the highest overall performance and was selected as the final model.

Phase 5: Feature Importance

Objective

Identify the most influential features affecting customer churn.

Top Features

- Contract
- Tenure
- Monthly Charges
- Total Charges
- Internet Service
- Tech Support
- Online Security
- Payment Method
- Paperless Billing
- Dependents

Observation

Customers with short contracts, high monthly charges, and fewer support services are more likely to churn.

Phase 6: Model Saving

Objective

Save the trained model for future predictions without retraining.

Steps Performed

- Saved the trained Gradient Boosting model using Joblib.
- Saved the StandardScaler object.
- Loaded both files in the Streamlit application for prediction.

Saved Files

- churn_model.pkl
- scaler.pkl

Phase 7: Streamlit Application

Objective

Deploy the trained model using Streamlit to create a simple and interactive prediction interface.

Features

- User-friendly interface.
- Customer information input form.
- Churn prediction.
- Prediction probability.
- Risk level (Low / Medium / High).
- Business recommendations based on prediction.

Benefits

- Easy to use.
- Real-time prediction.
- Helps telecom companies identify customers at risk of leaving.

Final Conclusion

This project successfully developed an end-to-end Machine Learning solution for predicting customer churn. After comparing three classification models, **Gradient Boosting** was selected as the best-performing model with **81% accuracy**. The model was deployed using Streamlit, allowing users to enter customer information and receive instant predictions along with churn probability and business recommendations. This system can help telecom companies improve customer retention by identifying high-risk customers early.



Customer Churn

Model Loaded Successfully

Developer

Ayesha Imran

Machine Learning Model

Gradient Boosting Classifier

Accuracy

81%

Dataset

Telco Customer Churn



Customer Churn Prediction System

Predict whether a telecom customer is likely to leave the company using Machine Learning.

Model

Gradient Boosting

Accuracy

81%

Dataset

7043 Rows

Features

19



Personal Information

Gender

Male

Monthly Charges

55.00

Senior Citizen

0

Total Charges

3300.00

Partner

Yes

Dependents

Yes

Tenure (Months)

25



Service Information

Phone Service

Yes

Device Protection

Yes

Multiple Lines

Yes

Tech Support

Yes

Internet Service

DSL

Streaming TV

Yes

Online Security

Yes

Streaming Movies

Yes

Online Backup

Yes



Billing Information

Contract

Two year

Payment Method

Bank transfer (automatic)

Paperless Billing

Yes

Predict Customer Churn

Prediction Result

✓ Customer is NOT likely to Churn

Churn Probability: 2.22%

Risk Level

● Low Risk

Business Recommendation

- 😊 Customer is likely to stay
- 💙 Maintain good service quality
- 🎁 Offer loyalty rewards

Customer Churn Prediction System

Developed by Ayesha Imran

Built with ❤️ using Python, Scikit-Learn & Streamlit